MITLE Math Camp Module 3: Probability and Statistics

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1 Probability

1.1 Elements of Set Theory

A simple framework for thinking about probability is by using set theory. A set is simply a collection of objects which we call elements. We denote a set using a letter like A and define $|A|$ as the number of elements in that set.

Example 1.1. *We can collect the ages of students for some fictional classroom in a set* $A = \{21, 25, 30, 25, 22, 20, 27, 25, 24, 27\}$. *The length of this set is 10.*

Several useful set operations can be used for multiple sets A, B, C

- (union) $A \cup B$: all elements in A OR B
- (intersection) $A \cap B$: all elements in both A AND B
- (difference) $A \setminus B$: all elements in A except those that are also in B

1.2 Sample space and events

A sample space for an experiment, denoted Ω, is a collection of possible outcomes from that experiment.

Example 1.2. *Consider the experiment of selecting 1 student at random from the fictional classroom and asking their age. In this case,* Ω *is defined as the set* $A = \{21, 25, 30, \ldots\}$ *from Example 1.1.*

*__Disclaimer:__ these notes are brief and far from comprehensive. Additional references can be found in the module outline shared separately as a Google Doc. Feel free to reach out to boysel@usc.edu with any questions or concerns.

1.3 Probability as a measure

Let A be an event defined in the sample space Ω. The probability of A is simply the number of possible outcomes that satisfy the conditions of A divided by the total number of possible outcomes in the sample space. In the simple case where Ω is finite:

$$P(A) = \frac{|A|}{|\Omega|}$$

Verify the following examples by simply counting the number of outcomes (e.g. draws from the classroom age list) that satisfy each event.

Example 1.3. *What is the probability of selecting a student who is 20 years old is*

$$P(\textit{student selected is 20}) = \frac{1}{10}$$

Example 1.4. *What is the probability of selecting a student who is 25 or 27?*

$$P(\textit{student is 25} \cup \textit{student is 27}) = \frac{3+2}{10}$$

Example 1.5. *What is the probability of selecting a student who is 25 or older?*

$$P(\textit{student selected older than 25}) = \frac{6}{10}$$

1.4 Counting

k mutually exclusive and collectively exhaustive events in n trials (Example: size of sample space when you flip a 2-sided coin n times)

$$k^n$$

$\{k_1, k_2, k_3, \ldots, k_n\}$ possible outcomes over the n trials (Example: license plate combinations)

$$\prod_{i=1}^{n} k_i$$

Number of ways n items can be arranged in order

$$n!$$

(Permutations) Count the number of ways to arrange k objects from a set of n total objects, where the *order of the selected objects matters.*

$$P(k, n) = \frac{n!}{(n-k)!}$$

Example 1.6. *Suppose there is a race with 5 runners. How many different top 3 finishes are possible?*

$$P(3, 5) = \frac{5!}{(5-3)!} = \frac{5!}{2!} = 5 \times 4 \times 3 = 60$$

(Combinations) Count the number of ways to draw k objects from a set of n total objects, where the *order of the selected objects does not matter.* note that since order doesn't matter, $P(k, n) \geq C(k, n)$ for any k, n.

$$C(k, n) = \frac{n!}{k!(n-k)!}$$

Example 1.7. *Suppose a class of 10 students must form pairs for an assignment. How many different pairings are possible (not that the context here gives us a clue that order does not matter, as it did in the race example above.).*

$$C(2, 10) = \frac{10!}{2!(10-2)!} = \frac{10!}{2!8!} = (10 \times 9)/2 = 45$$

1.5 Independent and mutually exclusive events

Let A and B be arbitrary events for some sample space Ω. Events A and B are said to be independent if the following (equivalent) statements hold

$$P(A \cap B) = P(A)P(B)$$
$$P(A \mid B) = P(A)$$
$$P(B \mid A) = P(B)$$

Events A and B are said to be mutually exclusive or disjoint if the following holds

$$P(A \cap B) = 0$$

In other words, A and B cannot occur at the same time. For example, if A is the event a flipped coin lands heads and B is the event it lands tails. In the single coin flip example, events A and B form a partitition of the sample space since $A \cap B = \Omega$. Each event in a partition (a set of events) is mututally exclusive and collectively exhaustive (i.e. the events *span* the sample space).

1.6 Laws of Probability, conditional probability, Bayes Rule

Let A and B be arbitrary events for some sample space Ω. Let $P(A \mid B)$ denote the conditional probability of A occurring given B also occurs. The following rules apply to any arbitray events such as A and B:

1. $P(A \cap B) = P(A)P(B \mid A)$
2. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
3. (Bayes rule)
$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$
   Note that we can derive Bayes' Rule from the formula in 1. (try this yourself)

Now assume the set of events $\{B_1, \ldots, B_n\}$ form a partition of Ω. We can then express $P(A)$ using the law of total probability:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i)$$

1.7 Random Variables

A useful modelling apprach for random experiments (e.g. doing empirical work) is to denote random and *ex ante* unknown quantities as random variables. A random variable, commonly denoted X, maps from the set of all possible outcomes in Ω to a number (e.g. $\mathbb{R}$. The point of all this is to make outcomes in some arbitrary outcome space measureable and quantifiable, eventually allowing us to model and estimate the likelihood of events. We distinguish between the random variable itself (random and therefore unknown) and the realized values it takes on during an experiment. We usually denote the realized value of X as x. This is best seen with examples.

Example 1.8. *Consider again the simple experiment of a single coin flip. The sample space is $\Omega = \{H, T\}$ where H denotes the event that the coin lands heads, T for tails. Now assume you didn't know the actual likelihood that the coin lands heads and you wanted to model it. Let X be a random variable that equals 1 if the coin lands heads and 0 if it lands tails*

$$X = \begin{cases} 1, & \textit{if } H \\ 0, & \textit{if } T \end{cases} \tag{1}$$

Suppose you next flip the coin 1000 times, recording the realized value of X each time, 1 for heads and 0 for tails, as x_i for $i = 1, \dots, 1000$. We can now estimate the probability of the event H (coin lands heads) in the following way:

$$P(\textit{coin lands heads}) = P(H) = P(X = 1) \approx \frac{\sum_{i=1}^{1000} 1\{x_i = 1\}}{1000}$$

Even though this example is somewhat trivial, the point of random variables is to model an arbitrary experiment in measureable terms so that likelihoods (i.e. probabilities) can be estimated.

1.8 Expected Value

Expected value is roughly equivalent to the probability weighted *average* of a random variable. We hinted at this in the random variable example with the coin flip in the previous section. In general (at least for the finite case), suppose a random variable X can take on values $x_1, \dots, x_n$ each with an corresponding probability $p_1, \dots, p_n$. The expected values of X, denoted $E[X]$, is defined

$$E[X] = \sum_{i=1}^{n} p_i x_i = \sum_{i=1}^{n} P(X = x_i) x_i$$

Example 1.9. *Let X denote the random variable associated with the single roll of a 6-sided die. The realized value of X is the number rolled (e.g. $\Omega = \{1, 2, 3, 4, 5, 6\}$). What is the expected value of X? Assume we know that the die is fair and that each number is equally likely. Using our counting rule, or conventional wisdom, we can determine that each outcome occurs with probabilty $1/6$. The expected value of X is therefore the sum of each outcome weighted by the probability it occurs, $1/6$*

$$E[X] = 1 \times (1/6) + 2 \times (1/6) + 3 \times (1/6) + 4 \times (1/6) + 5 \times (1/6) + 6 \times (1/6) = 3.5$$

1.9 Commonly used distributions

I will refer the reader to the references for commonly used distributions (in particular, the Hansen text on probability).

2 Statistics

2.1 Basics

We use statistics to estimate something we cannot completely observe. For example, suppose we wanted to know the average height of all humans on Earth. With infinite resources or technology, we could measure everyone and know this *population parameter*. Since we lack both, a cheaper alternative would be to take a subset of people and measure them to get a *sample estimate* of the true average human height. Much of statistics is concerned with estimating population parameters (fixed but unknown) by using sample estimate (random and not necessarily correct but known).

Example 2.1. *The sample mean* $\overline{x}_n$ *of a set of realizations* $\{x_1, \ldots, x_n\}$ *is defined* $\overline{x}_n = (1/n)\sum_{i=1}^{n} x_i$. *The sample mean is an estimator of the population mean* μ. *Similarly, the sample variance* s_n^2 *is defined* $s_n^2 = (1/(n-1))\sum_{i=1}^{n}(x_i - \overline{x}_n)$. *The sample variance is an estimator of the population variance* σ^2.

2.2 Sampling

If our goal is to estimate a population parameter, we want a sample that is representative of the larger population. If you wanted to know the average height for the entire world, a sample from a region with very short or very tall people would lead to a biased estimate of the population parameter. For most cases, we prefer a *random sample*, were each member of the sample is drawn with equal likelihood from the population. Note that we can sample with or without replacement. When the population is large, there is little difference between sampling with or without replacement since the likelihood of multiple selection is low.

2.3 Law of Large Numbers

In addition to being randomly drawn, many statistical applications benefit from samples being large. This is in part to leverage the Law of Large Numbers, which roughly states that the sample estimate of a population parameter becomes closer and closer to the true population parameter when the sample grows larger. We can also build intuition for the LLN by thinking about not adding more draws to the sample but removing them. Returning to the example of estimating a global mean height, consider the extreme case in which the entire population is the sample. In this case, the sample estimate of human height is exactly equal to the population parameter. Next, consider removing one person from the sample. The sample estimate in this case is less accurate and may not equal the population parameter as more and more observations are removed. Formally, if the sample estimate of n observations for a population parameter μ is denoted $\overline{X}_n$, then $\overline{X}_n$ converges in value to μ as n grows larger:

$$\overline{X}_n \to \mu \quad \text{as} \quad n \to \infty$$

2.4 Central Limit Theorem

In addition to the LLN, a more powerful result is the Central Limit Theorem, which roughly states that the sum from n random draws is normally distributed (a convenient, *well behaved* distribution) as n grows large. Formally, if $\{X_1, \ldots, X_n\}$ is a set of independent and identically distributed random variables (i.e. a random sample) with sample mean $\overline{X}_n = (1/n)\sum_{i=1}^{n} X_i$ and population mean μ and population variance σ^2, then the quantity $\sqrt{n}(\overline{X}_n - \mu)$ converges in distribution to $\mathcal{N}(0, \sigma^2)$ (i.e. a normal distribution with mean 0 and variance σ^2). Notice how the Central Limit Theorem pulls in many of the other concepts covered in this note: random variables, random sampling, expected values, the LLN. Among many other things, this is helpful for making statistical inference over sample estimates, allowing the researcher to make statements about the quantity they are estimating. In particular, hypothesis testing and confidence intervals will often make use of the CLT.

2.5 Hypothesis Testing

A primary method of statistical inference is hypothesis testing, using the realizations of random variables (i.e. data). Under the right conditions, a hypothesis test is used to make a statement about some unknown population parameter θ given data on the realizations of an experiment $S = \{x_1, \ldots, x_n\}$. A hypothesis compares a null hypothesis H_0 against an alternative hypothesis H_a. We initially assume that the null hypothesis is true and use the sample S to determine if we have enough evidence to reject the null hypothesis.

Example 2.2. *Suppose you have a sample $\{x_1, \ldots, x_n\}$ of the heights of n students from a school. Suppose you would like to estimate the claim that the average height in the entire school population is 1.7018 meters (i.e. $\mu = 1.7018$). In this case, the null hypothesis is that $\mu = 1.7018$ and the alternative hypothesis is $\mu \neq 1.7018$.*

$$H_0 : \mu = 1.7018 \quad H_a : \mu \neq 1.7018$$

An alternative way of thinking about this test is given the realized observation, do we have sufficient evidence in a statistical sense to reject the claim that the average height in the *entire school is 1.7018?*

Note that the example above involves a disjoint null and alternative (if one is true, then the other is false and vice versa). This is called a 2-sided test. To perform the test, we first form a test statistic:

$$\hat{t} = \frac{\overline{x}_n - 1.7018}{s/\sqrt{n}}$$

where s is the sample standard deviation. There are many possible test statistics so you must make sure you are choosing the correct one for the context. The one chosen above is known as a t-test since the population variance is unknown and n is assumed to be large. Under these assumptions, the test statistic follows a t distribution with $n-1$ degrees of freedom. This distributional assumption is what allows for inference (we are using the Central Limit Theorem here). If $\hat{t}$ is very large, then we have evidence that the sample estimate differs significantly from the population mean under the null (1.7018). Please see references for additional details (there are many).

2.6 Linear Regression

A workhorse tool in empirical analysis is the linear regression model. Linear regression models the relationship between random variables Y (outcome) and X (regressors). For observed data $\{y_i, x_i\}_{i=1}^n$, the simplest linear regression model can be expressed

$$y_i = \alpha + \beta x_i + \epsilon_i$$

where each outcome y_i is a function of a constant α, a coefficient β, the regressor x_i, and a unobserved "error" term ϵ_i. The observed data need not satisfy $y_i = \alpha + \beta x_i$ perfectly, hence the inclusion of an error term ϵ_i. With data in hand, we can *estimate* the coefficients (α, β) of the linear regression model. We interpret α as the level of Y when $X = 0$. We interpret β as the effect on Y when X is increased ($\frac{\partial Y}{\partial X} = \beta$). Since the errors ϵ_i cannot be observed, these are the only remaining unknowns. Estimation for the linear regression model is most often done using ordinary least squares (see references for details).

Example 2.3. *Suppose we are interested in the effect of capital investment X on a nation's GDP, Y. We can collect data from n countries on capital investment as $\{x_i\}_{i=1}^n$ and GDP as $\{y_i\}_{i=1}^n$. Here α is the estimated level of GDP when capital investment is 0. β is the estimated change in GDP when captial investment is marginally increased. If $\beta > 0$, then the relationship is positive.*

2.7 Estimating the gravity model of trade with linear regression

A pratical example of linear regression in action is estimating the so-called gravity model of international trade, which suggests that bilateral trade flows between nations is a function of both the size of a nation's economy as well as the distance between them. A simple version of the model is as follows:

$$Y_{ij} = G \cdot \frac{X_i X_j}{D_{ij}}$$

where Y_{ij} represents the volume of bilateral trade between nations i and j, X_i (X_j) is the size of i's (j's) economy, and D_{ij} is the distance between i and j. G is the "gravitational constant" (i.e. similar to α in the standard linear regression model notation). Note that this is a theoretical model and does not hold exactly. To estimate the model to estimation by linear regression, we need to adapt the model to include coefficients and an error term

$$Y_{ij} = G \cdot \frac{X_i^{\beta_1} X_j^{\beta_2}}{D_{ij}^{\beta_3}} \cdot \epsilon_{ij}$$

next, we can make this model linear by taking logs (i.e. log-linearization):

$$y_{ij} = g + \beta_1 x_i + \beta_2 x_j - \beta_3 d_{ij} + \epsilon_{ij}$$

where lowercase letters here represent the original quantities in logs (e.g. $y_{ij} = \log(Y_{ij})$). The model presented in the previous section involved only a single regressor. This model involves multiple regressors and is known as multiple linear regression. The interpretation of each coefficient is the effect on the outcome (left hand side) variable when the coefficient's corresponding regressor (e.g. x_i for β_1) is increased, *while holding the remaining regressors at their mean values* (e.g. x_j and d_{ij}). Using both conventional wisdom and empirical evidence from previous studies, we should expect the estimated coefficients in this model all to be positive: bilateral trade flows ought to be larger when the trading partners have larger economies and when they are closer together (i.e. when d_{ij} falls).